Java OOP Laboratory Exercise: Campus Café Ordering System

ITS 202: Object-Oriented Programming

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Section: C

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1. Introduction

The Campus Café Ordering System is a Java console program that records a customer's order of up to three menu items. It supports beverages, pastries, and meals. The program calculates item prices, applies a student discount or voucher, and awards loyalty points.

2. OOP Concepts Used

Classes and Objects

The program uses seven classes: MenuItem, Beverage, Pastry, Meal, Customer, Order, and CafeApp. CafeApp creates the objects used in the sample orders.

Abstraction

MenuItem is an abstract class. It declares the abstract computePrice() method so every menu-item subclass must provide its own pricing behavior.

Encapsulation

The classes keep their data in private or protected fields. Getters and setters control access, and setters reject invalid values such as a negative base price or negative number of extra sides.

Inheritance

Beverage, Pastry, and Meal extend MenuItem. They inherit common item information and reuse the parent class constructor.

Polymorphism and Method Overriding

The Order class stores MenuItem references. When it calls computePrice() or getDescription(), Java uses the version implemented by the actual object type.

Method Overloading

Customer has addPoints(int) and addPoints(double). Order also has computeTotal() and computeTotal(double), allowing a voucher to be applied.

Conditional Statements

if and if-else statements validate values, calculate size and warming fees, check the order limit, and apply the student discount.

Constructors

Default and parameterized constructors initialize menu items. Customer has three overloaded constructors for different customer details.

3. Program Demonstration

The first order is for Ana Reyes, a student. It contains an iced latte, warmed ensaymada, and upsized chicken adobo with one extra side. A fourth item is attempted to demonstrate the three-item limit. The second order is for Ben Cruz and demonstrates the default Beverage constructor, setter validation, and a ■20 voucher.

4. Challenges and Solutions

One challenge was organizing the program into separate classes while allowing Order to work with different menu-item types. This was solved by making MenuItem an abstract parent class and storing the items as MenuItem references. Another challenge was preventing invalid values. Validation was added to setters using conditional statements. The order limit was handled with three individual item fields, as required by the activity.

5. Program Output Screenshot

After compiling and running CafeApp, insert a screenshot of the actual console output in this section. The screenshot should show both receipts, the full-order message, and the negative-price validation message.

